

Exercise 13: The discretized Ekman Equations

1. Using forward Euler in time:

zonal: $\frac{U_k^{n+1} - U_k^n}{\Delta t} - f v_k^n = A_v \frac{U_{k+1}^n - 2U_k^n + U_{k-1}^n}{\Delta z^2}$

meridional: $\frac{U_k^{n+1} - V_k^n}{\Delta t} + f u_k^n = A_v \frac{V_{k+1}^n - 2V_k^n + V_{k-1}^n}{\Delta z^2}$

So: $U_k^{n+1} = U_k^n + \Delta t \left[f v_k^n + \frac{A_v}{\Delta z^2} (U_{k+1}^n - 2U_k^n + U_{k-1}^n) \right]$

$$V_k^{n+1} = V_k^n + \Delta t \left[-f u_k^n + \frac{A_v}{\Delta z^2} (V_{k+1}^n - 2V_k^n + V_{k-1}^n) \right]$$

2. $\frac{\partial W}{\partial t} + i f W = A_v \frac{\partial^2 W}{\partial z^2}$

$$W_k^{n+1} = W_k^n - i f \Delta t W_k^n + r (W_{k+1}^n - 2W_k^n + W_{k-1}^n);$$

$$r = \frac{A_v \Delta t}{\Delta z^2}$$

$$G = 1 - 4r \sin^2 \left(\frac{\kappa \Delta z}{2} \right)$$

$$r = \frac{A_v \Delta t}{\Delta z^2} \leq \frac{1}{2}$$

Used
Claude AI
for this one

measures how big a spreading jump you take per step

TWO REAL EQUATIONS

$$\begin{aligned} & \text{u-equation (E-W)} \\ & \partial u / \partial t - f v = A v \partial^2 u \\ & \text{v-equation (N-S)} \\ & \partial v / \partial t + f u = A v \partial^2 v \end{aligned}$$

combination of
two equations

ONE COMPLEX EQUATION

combine $\rightarrow$
($W = u + i v$)

$$\begin{aligned} & \partial W / \partial t + i f W = \\ & A v \partial^2 W / \partial z^2 \end{aligned}$$

discretise the 'now'
single equation

forward Euler in time,
centred in space (z)

$$\begin{aligned} W_k^{n+1} &= W_k^n - i f \Delta t \cdot W_k^n \\ &+ r (W_{k+1}^n - 2 W_k^n + W_{k-1}^n) \\ \text{where } r &= A v \Delta t / \Delta z^2 \end{aligned}$$

ask: does W_k^n stay bounded as n grows?

find the factor G

von Neumann $\rightarrow$ amplification factor G

stability needs $|G| \leq 1 \rightarrow r \leq \frac{1}{2}$

3. Vertical (diapycnal) eddy viscosity $= A_v = 10^{-4} \text{ m}^2 \text{ s}^{-1}$

$$\Delta t \leq \frac{\Delta z^2}{2A_v}$$

$$\Delta t \leq \frac{(2)^2}{2 \times 10^{-4}} = 2 \times 10^4 \text{ s} \approx 5.6 \text{ h}$$

- $\Delta z = 1.5 \text{ m}$
- $\Delta t = 180 \text{ s}$

my random choice

$$r = \frac{10^{-4} \cdot 180}{(1.5)^2} = 8 \times 10^{-3} = 0.008$$

- $\Delta z = 2 \text{ m}$
- $\Delta t = 160 \text{ s}$

from exercise notes

$$r = \frac{10^{-4} \cdot 160}{(2)^2} = 4 \times 10^{-3} = 0.004$$

both are stable, because both 0.008 and 0.004 are smaller than 0.5